

Generative AI and LLM Foundations

Topics

- e! What is Generative AI?
- e! Generative AI Evolution
- e! Differentiating Generative AI from Discriminative AI
- e! Types of Generative AI
- e! Generative AI Core Concepts
- e! LLM Modelling Steps
- e! Transformer Models: BERT, GPT, T5
- e! Training Process of an LLM Model like ChatGPT
- e! The Generative AI development lifecycle
- e! Overview of Proprietary and Open Source LLMs
- e! Overview of Popular Generative AI Tools and Platforms
- e! Ethical Considerations in Generative AI
- e! Bias in Generative AI outputs
- e! Safety and Responsible AI Practices

Learning Objectives

By the end of this module, you will be able to:

- e! Define generative AI and differentiate it from traditional AI models.
- e! Describe transformer models and explain attention mechanisms.
- e! Compare BERT, GPT, and T5 LLMs, highlighting their core functions.
- e! Outline the stages of LLM development, from pretraining to deployment.
- e! Identify generative AI tools for prototyping, tuning, and deployment workflows.
- e! Apply responsible AI principles to ensure fairness, safety, and transparency.

The Rise of Generative AI

Problem Statement: Riya is managing several critical tasks simultaneously and requires an effective solution to accomplish them within tight deadlines while maintaining high quality.

I leveraged Generative AI to draft content, analyze data, and structure tasks. It significantly improved my efficiency.



I thought AI only answered questions. Can it also create and structure work?

Introduction to Generative AI

What is Generative AI?

Generative AI is gaining attention for its ability to create human-like content, text, images, audio, and more.



Its rapid progress is transforming industries like media, healthcare, education, and customer service.

Unlike older automation tools, it can create new material from patterns it has learned.

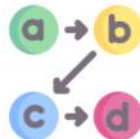
Generative AI: A Shift in Machine Intelligence

Generative AI refers to advanced systems that analyze existing data to generate original content, mimicking the characteristics of human-created work.

Key Shift



From analyzing data to creating content.



From rule-based logic to learning-based creativity.

Generative AI within the AI Landscape



01

Artificial Intelligence (AI)

The broader field focused on enabling machines to perform intelligent tasks.



02

Machine Learning (ML)

A subset of AI that enables machines to learn from data patterns.



03

Generative AI

A branch of ML that generates new data based on learned patterns.

It complements other AI approaches, such as predictive models and reinforcement learning.

Foundation Models: A Bridge to LLMs



Foundation Models: Large-scale models trained on diverse data, capable of performing multiple tasks.



LLMs: LLMs (like GPT) are foundation models for language understanding and generation.

They are called “foundations” because they form the base for more specific models.

Why Generative AI Matters Today

Makes human-AI interaction more natural and useful.



Enables creativity at scale (writing, designing, coding).

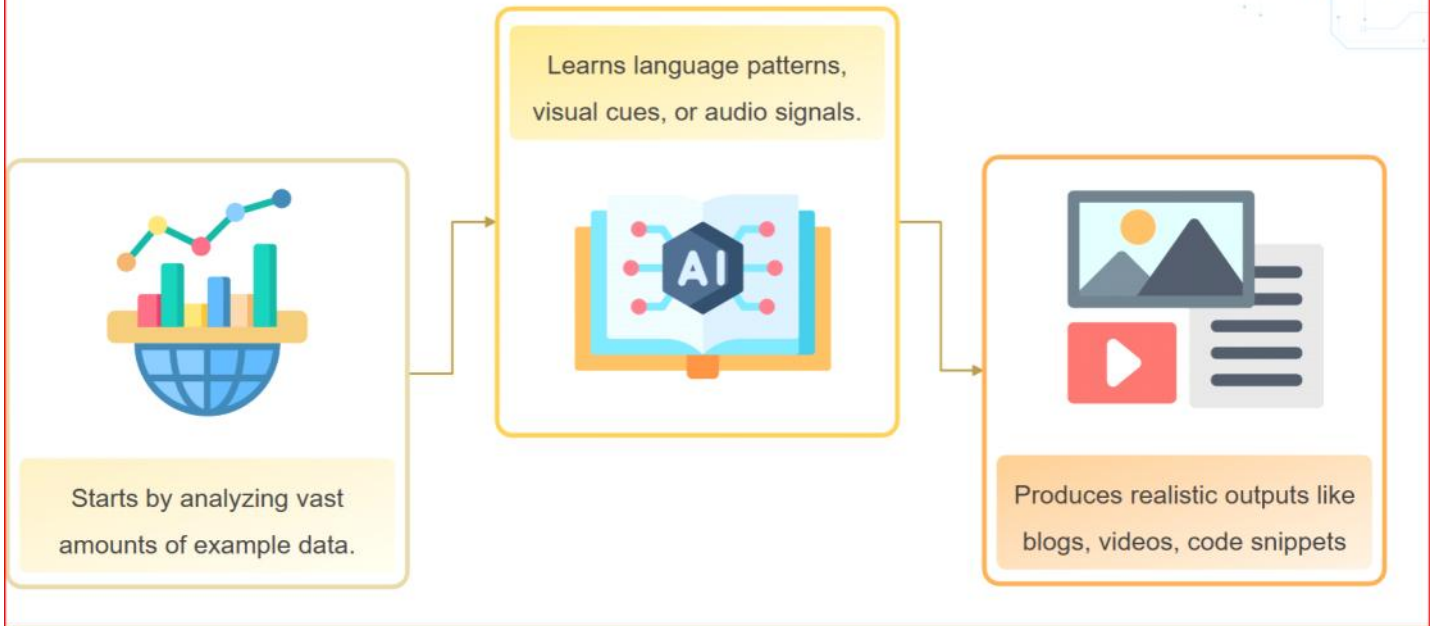


Boosts productivity by assisting with repetitive or complex content generation.

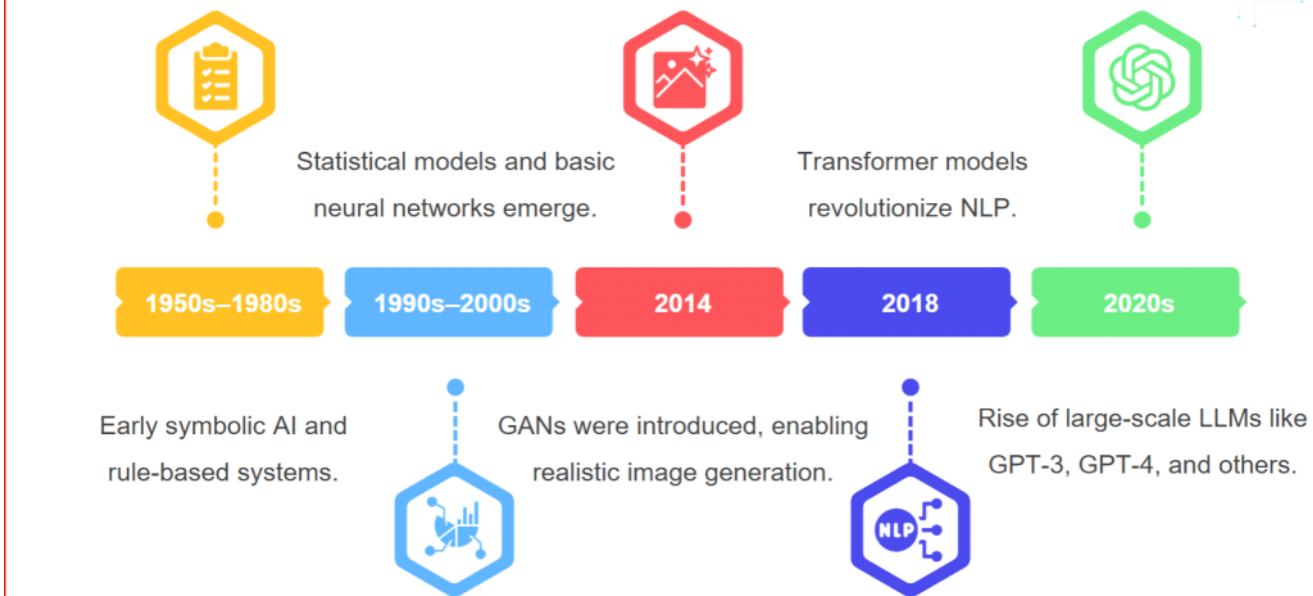


Democratizes access to skills through tools like chatbots and copilots.

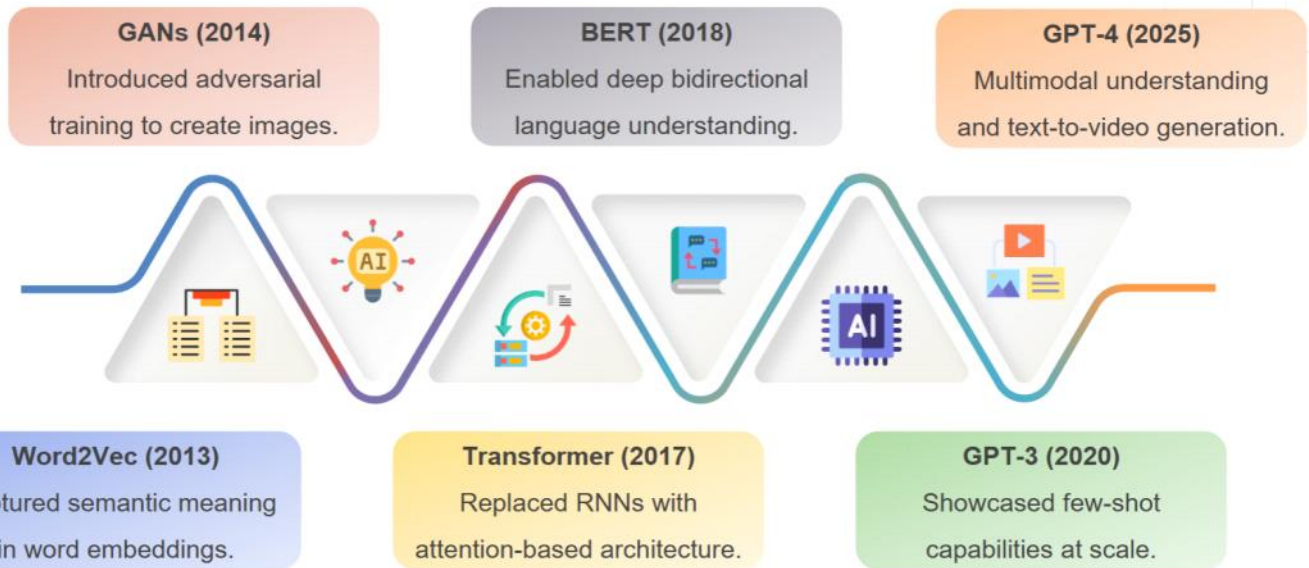
From Data to Content: Generative AI in Action



Timeline of Generative AI Progress



Landmark Models and Innovations



Real-World Applications



Rule-Based to Neural Generative Models



Rule-Based

Hardcoded logic with no learning ability.



Statistical Models

Relied on probability rules with limited creativity.



Neural Models

Learned directly from data to produce original outputs.

Neural approaches are now dominant due to flexibility and scalability.

Discriminative vs. Generative AI

Discriminative models

Discriminative models identify patterns to classify input.

Example: "Is this spam or not?"

Core equation: $P(y|x)$

Models the **conditional probability** of label y given input features x .



Generative models

Generative models learn how data is distributed and can produce new examples.

Example: "Write an email like this one."

Core Equation: $P(X|y)$

Models the **joint probability distribution** of inputs x and labels y together.



This distinction helps us choose the right tool for a specific task.

Core Idea: Generation vs. Classification

Generative AI: Learns to model how data is distributed and creates new examples.



Generation answers
"What can I create?"

Discriminative AI: Learns to separate or classify input into categories.



Discrimination answers
"What class does this belong to?"

Side-by-Side Model Examples

Task	Generative Model	Discriminative Model
Text generation	GPT and T5	Text classifier
Image creation	GANs and Diffusion Models	ResNet and MobileNet
Speech	AudioLM and WaveNet	Speech classifier

Model Choice Depends on Task: Generation or Classification

Choosing the Right AI Model



Generative AI

Produces new content

Handles incomplete inputs

More resource-intensive

Harder to train and evaluate



Discriminative AI

Fast and efficient

Accurate for classification

Cannot generate new data

Limited flexibility for open-ended tasks

GANs: Adversarial Generation Explained

Generative Adversarial Networks (GANs)

Two models in a contest



Generator: Tries to create realistic data.



Discriminator: Tries to detect fake from real.

Over time, the generator learns to fool the discriminator.

Applications: Image synthesis, face generation, style transfer.

VAEs and Diffusion Models

Variational Autoencoders (VAEs)



VAEs encode data into a compressed space and decode it to reconstruct the original data, making them effective for smooth and structured generation.

Diffusion Models

Diffusion models reverse a noise process, starting with noise to reconstruct data, achieving high-quality image generation.



Language Models as Generative Systems

Language models learn the context in which words appear.



Trained on vast datasets, they understand grammar, semantics, and style.



1

2

3

4



They predict the next word or phrase in a sentence or paragraph.



Examples: GPT, BERT (masked language modeling), T5 (text-to-text).

The Rise of LLMs as Universal Generators

They can summarize, translate, write code, and generate long-form content.

Few-shot and zero-shot prompting makes them flexible.

Large Language Models (LLMs) handle diverse tasks without task-specific tuning.

LLMs like GPT-4 are central to modern generative AI.



Generative AI Core Concepts

Tokenization and Input Representation

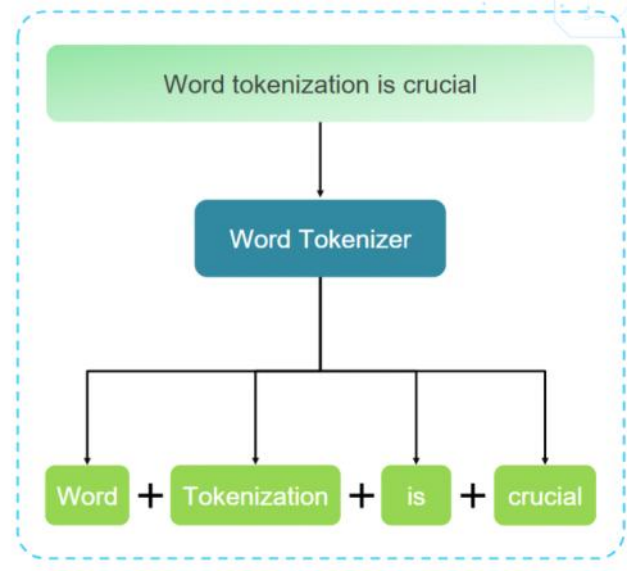


Tokenization: Converting text into smaller units (tokens) such as characters, sub-words, or words.

Tokens can be characters, sub-words, or words.

Helps models understand and process raw input consistently.

Common tools: Byte Pair Encoding (BPE) and WordPiece.



Word Embeddings and Semantic Meaning

Transforms tokens into vector representations.



Captures relationships like, $\text{King} - \text{Man} + \text{Woman} = \text{Queen}$

Embeddings help models grasp meaning and context.



Examples: Word2Vec, GloVe, contextual embeddings from transformers.

Language Modeling and Probability



Predicts the likelihood of the next word in a sequence.



Models assign probabilities to all possible next tokens.



Forms the core of generative text systems.



Example: "The cat sat on the ____"



high probability for **"mat"**.

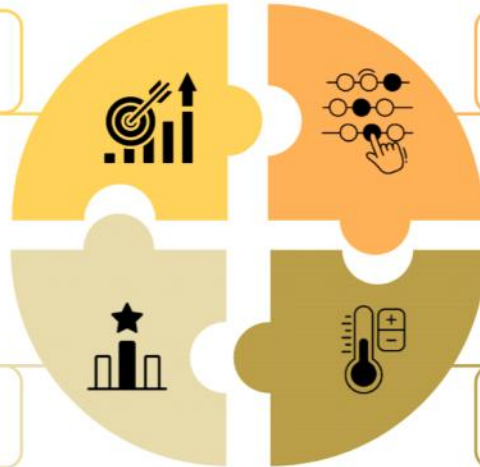
Decoding Strategies

Greedy decoding: Picks the highest probability word each time.

Top-p: Chooses from the smallest set of words whose total probability exceeds p .

Top-k: Chooses from the top k likely options.

Temperature: Controls randomness. Lower = focused, higher = creative.



Decoder-Only vs. Encoder-Decoder

Decoder-Only (Example: GPT)



Great for generation tasks.



Predicts the next token using only the past context.

Encoder-Decoder (Example: T5, BART)



The encoder processes input; the decoder generates output.



Better for translation, summarization, and Q&A tasks.

Transformer Models - BERT, GPT, and T5

What is a Transformer?

A transformer is a deep learning model architecture that relies entirely on attention mechanisms to process data in parallel, rather than sequentially.

Why it matters

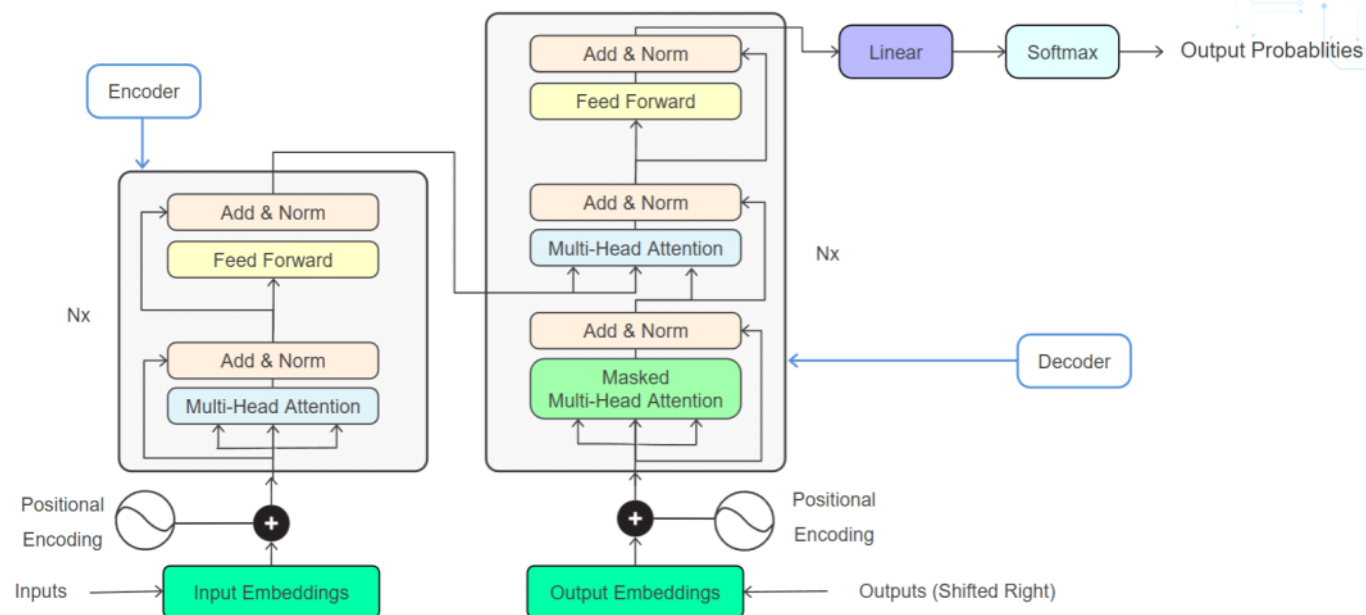
Replaced older models like RNNs and LSTMs.

Can handle long sequences efficiently.

Forms the backbone of modern language models like BERT, GPT, and T5.



Transformer Architecture Breakdown



Attention Mechanism: Focus in Context

Attention lets the model decide what words to focus on when generating or interpreting text.

Example: In "The animal didn't cross the road because it was too tired," attention helps the model link "it" to "animal."

it



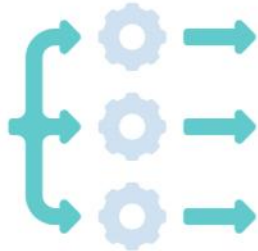
animal

Attention Score: 0.92

Scores are calculated to weigh the importance of words in context.



Multi-Head Attention: Diverse Learning



Instead of one attention calculation, transformers use several heads in parallel.



Each head focuses on a different aspect: syntax, coreference, and meaning.



Outputs are combined to form a rich understanding of the input.

BERT - Bidirectional Context Understanding

A pre-trained language model that understands word meaning by considering both preceding and following contexts.



BERT looks at words before and after a target word to understand its meaning in context.

01



Pretrained with Masked Language Modeling and Next Sentence Prediction.

02



Performs well on tasks like question answering, classification, and named entity recognition.

03

GPT: Autoregressive Generation in Action



GPT generates one token at a time, left to right.

Employs decoder blocks that operate unidirectionally.



Excellent at generating essays, stories, summaries, and code.

Learns to predict the next word from context.



T5: Text-to-Text Framework

Transforms all tasks into a text-to-text format.



Example

Input: "Translate English
to German: How are you?"



Output: "Wie geht es dir?"



Useful for summarization, translation, classification, and generation.

Positional Encoding in Transformers

Positional Encoding supplies the model with position information for each token, allowing transformers to capture word order despite processing inputs in parallel.

Since transformers don't process tokens in order, they need positional encodings.

Often uses sine and cosine functions for this encoding.



Encodes each token's position to help the model understand sequence order.

Crucial for understanding grammar and word order.

Comparing BERT, GPT, and T5 Architectures

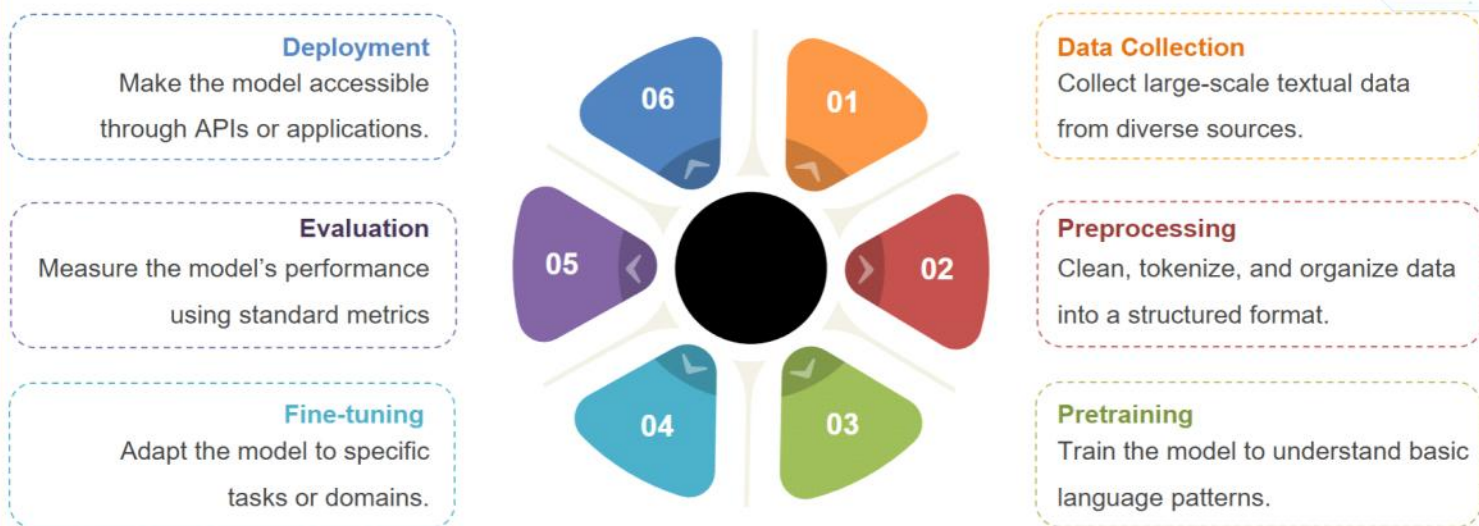
Model	Directionality	Architecture	Use Case Focus
BERT	Bidirectional	Encoder only	Understanding
GPT	Unidirectional	Decoder only	Generation
T5	Bidirectional	Encoder-Decoder	Flexible text tasks

Each is optimized for different language tasks.

Demo 1: Creating a Small Transformer using PyTorch

LLM Modelling Steps

LLM Pipeline Overview



Pretraining at Scale

Pretraining is done on huge corpora, books, web pages, research papers.



The goal is to learn grammar, reasoning, and factual knowledge.



Predict the next word (causal).

Uses self-supervised tasks



Predict missing words (masked).

Requires massive compute and training time.



RLHF: Human Feedback for Better Output

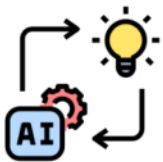
RLHF (Reinforcement Learning from Human Feedback) trains models by using human feedback to rank responses, helping the model generate more accurate, helpful, and safe outputs.



Combines AI outputs with human preferences to guide training.



Human reviewers rank responses, which trains a reward model.



Model learns to generate more helpful, honest, and harmless responses.



Widely used in ChatGPT and other assistants.

Scaling Laws: Bigger is Better?

Research shows that larger models often perform better.



Compute power



Data



Parameters

Accuracy improves with more

But growth comes with trade-offs

Cost



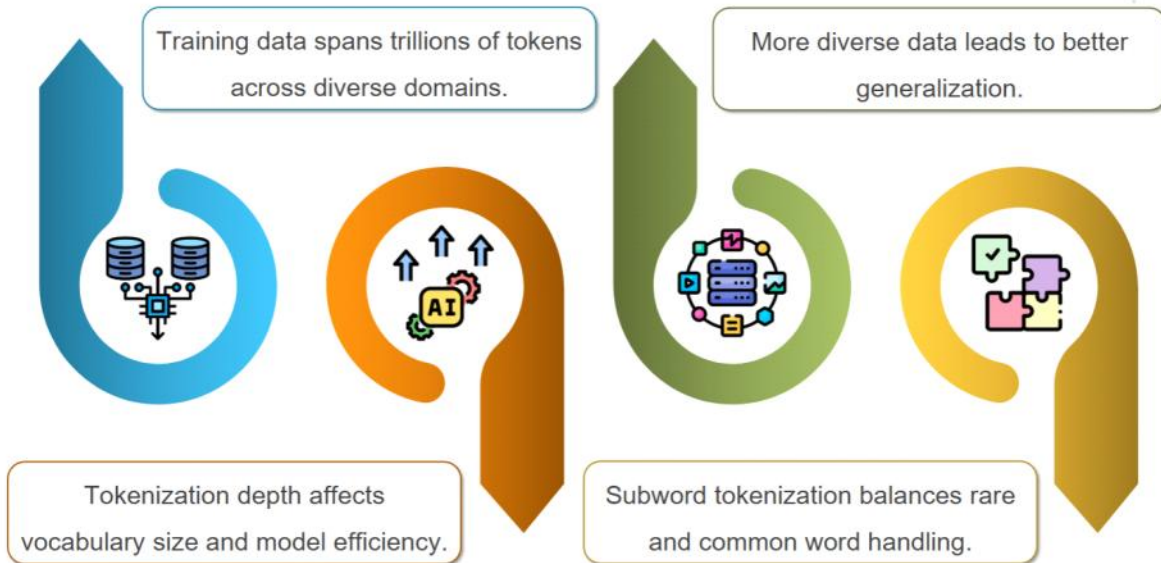
Latency



Energy use



Data Needs and Tokenization Depth



The Generative AI Development Lifecycle

Lifecycle Stages: Idea to Deployment

Stages in developing a generative AI system

Model Selection or Training

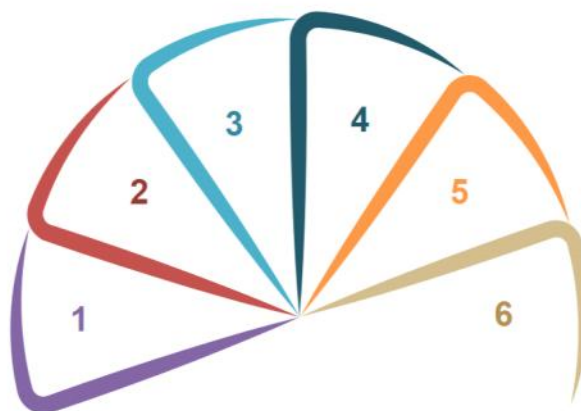
Choose a prebuilt model or train your own.

Data Preparation

Clean, label, and structure the training data.

Problem Definition

What task should the model help with?



Integration

Connect the model to your app, website, or process.

Monitoring

Track output quality and user interaction.

Iteration

Improve based on feedback and new needs.

Evaluating LLM Outputs



BLEU: Measures how close generated text is to a reference.



ROUGE: Focuses on recall, how much relevant content is captured.



Human Evaluation: Assesses clarity, fluency, accuracy, and relevance; most reliable but slow.

Model Drift and Continuous Improvement

Over time, a model's output may become less accurate or relevant.



Changes in real-world data

Causes of model drift

User behavior shifts



Regular retraining or prompt adjustments help reduce drift.



Monitoring is essential for long-term quality.

Managing Model Versions

Use version control to track



Prompt changes



Fine-tuned model updates



Dataset revisions

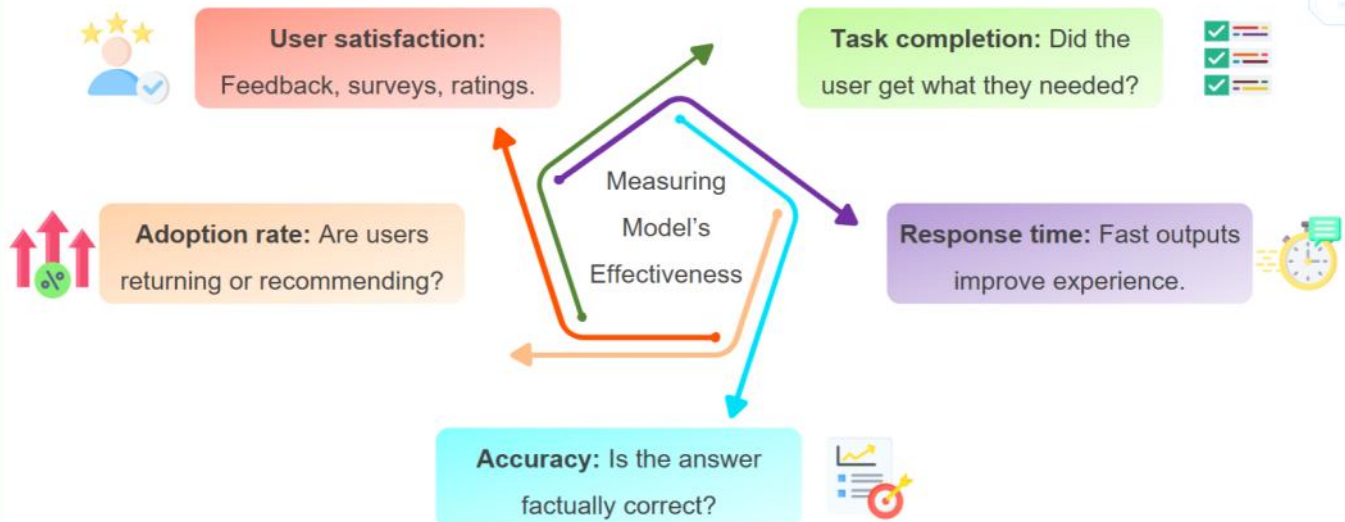


Helps roll back to stable versions and audit performance over time.



Naming conventions and changelogs are useful for teams.

Success Metrics for LLM Use Cases



Overview of Proprietary and Open Source LLMs

Proprietary vs. Open Source LLMs

Category	Proprietary	Open Source
Ownership	Closed, company-owned	Community or organization maintained
Access	API-based	Downloadable and modifiable
Examples	GPT-4, Claude, and Gemini	LLaMA, Mistral, and Falcon
Customization	Limited	Highly flexible
Cost	Usage-based pricing	Mostly free to run, compute cost applies

Popular LLM Models



GPT-4 (OpenAI): Has general capability, strong reasoning, and code generation.



Claude (Anthropic): Focuses on helpfulness, safety, long context.



Falcon: Scalable open models suited for a wide range of NLP tasks.



Gemini (Google DeepMind): Multimodal, highly integrated with Google products.



Mistral: Lightweight, fast models optimized for on-device or cloud use.



LLaMA (Meta): Open source, strong performance with smaller model sizes.

Use Case Fit: Choosing the Right LLM

Enterprise apps

GPT-4 or Claude (stable APIs, compliance-ready).



Budget-sensitive teams

Falcon or Mistral (open, efficient).

Multilingual tasks

Gemini and LLaMA show strong results.



Experimentation or control

Open-source Models are ideal for customizable support

Popular Generative AI Tools and Platforms

Popular Generative AI Tools



OpenAI Playground

Test and refine prompts using GPT models.

Simple interface for trying different instructions, tones, and completions.



Hugging Face

A hub for open source models.

Offers model hosting, fine-tuning, and APIs.



watsonx.ai (IBM)

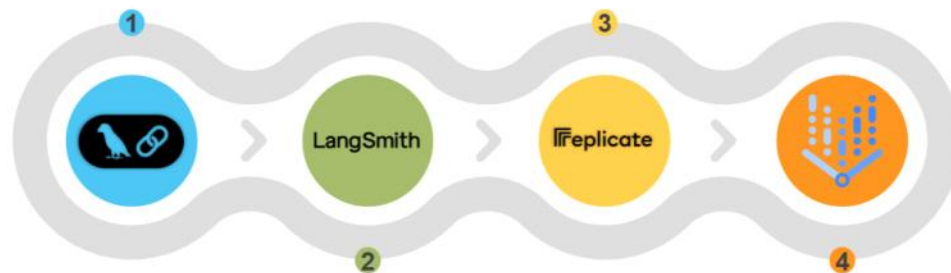
Enterprise-ready tool focused on transparency and governance.

Supports both foundation and domain-specific models.

Prototyping and Deployment Platforms

LangChain: Creates apps with chained prompts and memory.

Replicate: Hosts and runs models via simple API calls.



LangSmith / Google AI Studio: Debug, monitor, and prototype LLM and Gemini apps.

Azure OpenAI / Vertex AI: Enterprise-grade deployment with security, scaling, and monitoring.

Best-Fit Tools for Beginners and Enterprises

User Type	Recommended Tools	Why It Works
Beginners	OpenAI Playground, Hugging Face	No setup needed, simple to explore
Developers	LangChain and Replicate	Easy APIs and integration ready
Enterprises	watsonx.ai, Azure OpenAI, Vertex AI	Governance, compliance, scalability

Ethical Considerations and Responsible AI

Bias, Safety, and Explainability in GenAI

Key concerns with AI-generated content:



Bias: Models can reflect social or cultural biases from training data.



Safety: Potential to generate toxic, harmful, or misleading responses.



Explainability: Outputs are hard to interpret or trace back to logic.

Trust, fairness, and public acceptance depend on addressing these issues.

Real-World Failures and Lessons

Examples of what can go wrong



Chatbots generating offensive or harmful content when prompted.



Fake news creation using AI-generated articles or voices.



Misleading medical or legal advice from unchecked models.

Lesson: Even high-performing models need safeguards and oversight.

Responsible GenAI Principles

Test for harmful outputs during development.

Disclose AI-generated content clearly to users.

Use prompt filters, content moderation, and human-in-the-loop systems.

Ensure accessibility and reduce bias across user demographics.

Regulations and Societal Impact

Governments and global bodies are working on AI-specific laws.

Areas of regulation include



Data privacy

Deepfake detection



AI disclosures

Responsible developers must



Implement ethical guidelines in model development.

Ensure user consent and informed usage



Demo 2: Explore OpenAI Playground to test text generation

Summary

In this lesson, you have learned to:

- e! Define Generative AI and how it creates new content.
- e! Explain transformer architecture and the role of attention mechanisms.
- e! Distinguish between BERT, GPT, and T5 model capabilities.
- e! Identify key stages in building and deploying large language models.
- e! Explore GenAI development tools, such as Hugging Face and LangChain.
- e! Recognize responsible AI principles, including bias, safety, and explainability.